

		rocket.
10	A	For one molecule, elastic collision with the shaded wall results in a change of momentum $-2mv$ (taking rightward as positive). The molecule has to travel a distance $2p$ before it next collides with the same shaded wall. Hence, the time between collisions with the same wall is $\Delta t = \frac{2p}{v}.$ The rate of change of momentum of this molecule due to collision with the shaded wall is $\frac{\Delta p}{\Delta t} = \frac{-2mv}{\frac{2p}{v}} = \frac{-mv^2}{p}.$ By Newton's second law of motion, the force that the wall exerts on the molecule is $F_{\text{on molecule}} = \frac{\Delta p}{\Delta t} = \frac{-mv^2}{p}.$ The negative sign means that the force acting on the molecule is pointing away from the shaded wall. By Newton's third law, the force acting on the wall by the molecule is $F_{\text{on wall}} = -F_{\text{on molecule}} = \frac{mv^2}{p}.$

Qn	Answer	Solution
		For N molecules that collide elastically with the shaded wall with velocities v and rebound, the average force acting on the shaded wall is $F_{\text{average}} = \underbrace{\frac{mv^2}{p} + \frac{mv^2}{p} + \dots + \frac{mv^2}{p}}_N = \frac{Nmv^2}{p} = \frac{mNv^2}{p}.$
11	B	At 0 °C, internal energy of ideal gas